

## Preliminary Exam in Analysis, August 2025

**Problem 1.** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a continuous function, and define

$$g(x) = \sum_{n=1}^{\infty} \left( \frac{|f(x + \frac{1}{n})|}{1 + |f(x + \frac{1}{n})|} \right)^n.$$

Prove that  $g$  is a continuous function on  $\mathbb{R}$ .

**Problem 2.** Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$  be continuous, and let  $K \subset \mathbb{R}^n$  be compact. Assume that  $f$  is one-to-one on  $K$  and also  $f$  is one-to-one in a neighborhood (in  $\mathbb{R}^n$ ) of every  $x \in K$ . Prove that there is an open set  $U \supset K$  such that  $f$  is one-to-one in  $U$ .

**Problem 3.** Assume that  $f \in C^1(\mathbb{R}^2)$  is such that

$$\frac{\partial f}{\partial x}(x, y) = \frac{\partial f}{\partial y}(x, y) \quad \text{and} \quad f(x, 0) > 0 \quad \text{for all } x, y \in \mathbb{R}.$$

Prove that  $f(x, y) > 0$  for all  $x, y \in \mathbb{R}$ .

**Problem 4.** Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a function such that  $g : \mathbb{R}^2 \rightarrow \mathbb{R}$  defined by

$$g(u, v) := f(e^u \cos v, e^u \sin v)$$

has continuous partial derivatives  $\partial g / \partial u$  and  $\partial g / \partial v$ . Prove that  $f$  is differentiable on  $\mathbb{R}^2 \setminus \{(0, 0)\}$ .

**Problem 5.** Given a positive integer  $n$ , prove that there exists  $\varepsilon > 0$  such that for every  $n \times n$  matrix  $A$  satisfying  $\|A - 2I_n\| < \varepsilon$ , there exists an  $n \times n$  matrix  $B$  such that  $A = B^3 + B^T$ , where  $B^T$  denotes the transpose of  $B$ ,  $I_n$  is the  $n \times n$  identity matrix, and for  $A = (a_{ij})_{1 \leq i, j \leq n}$ ,

$$\|A\| = \left( \sum_{i=1}^n \sum_{j=1}^n a_{ij}^2 \right)^{1/2}.$$

**Problem 6.** Let  $x = (x_1, x_2, x_3)$ ,  $\mathbf{0} = (0, 0, 0)$ , and  $B_r = \{x \in \mathbb{R}^3 : |x| \leq r\}$ . Let  $f : \mathbb{R}^3 \setminus \{\mathbf{0}\} \rightarrow \mathbb{R}$  be a  $C^1$  function such that  $|f(x)| \leq |x|^{-3/2}$  for all  $x \neq \mathbf{0}$ . Prove that, for every  $a > 0$ ,

$$\lim_{\delta \rightarrow 0^+} \iiint_{B_a \setminus B_\delta} [x \cdot \nabla f(x) + 2f(x)] \frac{dV}{|x|} = \iint_{|x|=a} f(x) dS.$$